ENTERPRISE ARCHITECTURE MANUAL

Marketplace Payouts, Razorpay Settlement Cycles & Zero-Balance Cancellation Architecture

A rigorous architectural and operational manual detailing the end-to-end payment lifecycle: 90/10 automated escrow splits, scheduled banking settlement rails, zero-balance negative auto-offsets, and the Turfify 4-Tier dynamic cancellation matrix.

RAZORPAY ROUTE 90/10 | TDS SEC 194-O (1%) | ZERO-BALANCE AUTO-OFFSET | IN-APP WALLET CREDITS

PLATFORM STACK	MARKETPLACE SPLIT	SETTLEMENT MODE	CANCELLATION MATRIX
Razorpay + Route	90% Owner / 10% App	T+1 / Escrow Hold	4-Tier Dynamic

📌 **The Primary Operational Dilemma Answered in This Document:**

"A customer books on Wednesday for a Saturday slot (₹1,500). By Friday, Razorpay has already settled the payout into the turf owner's bank account, leaving the Razorpay gateway balance at ₹0. When the customer cancels on Saturday, how does Turfify issue the refund without incurring out-of-pocket deficit or cash flow disruption?"

This document resolves this query across three layers: (1) Razorpay's native negative-balance auto-deduction rail, (2) the venue cancellation policy (why Saturday is non-refundable), and (3) Turfify In-App Wallet credits providing instant ₹0-cost resolution.

Architectural Pillars of the Turfify Payment Engine

1. Automated Marketplace Splits (Razorpay Route)

Zero manual banking transfers. The moment a ₹1,500 booking is authorized, Razorpay Route programmatically carves 90% (₹1,335 net after 1% TDS) to the turf owner's linked bank account and retains 10% (₹150) in Turfify's platform treasury.

2. Zero-Balance Negative Offset Mechanism

If a refund is authorized when the merchant balance is ₹0, Razorpay automatically creates a negative merchant balance (e.g. -₹1,500) and reconciles it against incoming weekend transactions automatically, eliminating manual replenishment.

3. Time-Decayed 4-Tier Cancellation Matrix

Protects turf owners from catastrophic last-minute slot abandonment. Cancellations made within 6 hours of kickoff are strictly non-refundable (0%), ensuring owners keep their earnings even if funds have already settled.

4. In-App Wallet Credit Optimization (₹0 Gateway Cost)

Encourages players to take refunds as 100% Turfify Wallet credits. Cash never leaves the platform, zero payment gateway fees are burned, and 100% of customer spending remains locked in the Turfify sports ecosystem.

The End-to-End Payout & Settlement Flow (Wednesday to Saturday)

A complete chronological trace of a ₹1,500 booking: from player UPI checkout on Wednesday to bank settlement on Friday, illustrating exactly how and when funds migrate between escrow and vendor accounts.

1

Day 0 (Wednesday 14:00) — Player Checkout & Payment Authorization

PAYMENT.CAPTURED

Player selects Saturday 20:00–21:00 slot. Backend creates a Razorpay Order with programmatic Route transfers. Player authorizes ₹1,500 via UPI Intent (GPay/PhonePe). Razorpay captures the payment and fires an idempotent webhook to the Turfify server.

2

Day 0 (Wednesday 14:02) — Razorpay Route Marketplace Split & Tax Escrow

90% / 10% SPLIT

Razorpay Route automatically splits the transaction inside the gateway:

- **10% Platform Fee:** ₹150.00 retained in Turfify platform balance.
- **1% Section 194-O TDS:** ₹15.00 auto-withheld for quarterly government e-filing (Form 16A).
- **90% Owner Net Share:** ₹1,335.00 credited to the turf owner's linked nodal account (`acc_owner123`).

3

Day 1 to 2 (Thursday 04:00 AM – Friday) — Automated Bank Settlement

SETTLEMENT T+1

Under standard Razorpay T+1 / T+2 settlement cycles, funds accumulated in the owner's linked account are swept via automated NEFT/IMPS directly to their commercial bank account (e.g. HDFC, ICICI, SBI).

4

Day 3 (Saturday Morning) — The Balance Status: Gateway Reaches ₹0.00

BALANCE = ₹0.00

Because the automated settlement completed on Friday morning, the turf owner now has the ₹1,335 cash safely in their private bank account, and the Razorpay merchant wallet balance is currently **₹0.00**.

Financial Breakdown of a Single ₹1,500 Booking:

GROSS BOOKING	TURFIFY TAKE-RATE (10%)	TDS SEC 194-O (1%)	OWNER PAYOUT (NET)
₹1,500.00	**₹150.00**	**₹15.00**	**₹1,335.00**
Paid via UPI / Cards	Gross Revenue	Automated Tax Deposit	Settled Friday 04:00 AM

Settlement Schedule Strategies: T+1 Standard vs Hold-Until-Kickoff

SETTLEMENT STRATEGY	WHEN MONEY HITS BANK	CANCELLATION IMPACT	TURFIFY RECOMMENDATION
Standard T+1 Cycle	Next business day (Thursday/Friday)	Gateway balance reaches ₹0 before game day. Requires negative balance offset or wallet credit.	GATEWAY DEFAULT
Escrow Hold Settlement (`on_hold: 1`)	Sunday morning (T+1 *after* the match is played)	100% of funds remain in Razorpay escrow until kickoff. Zero balance deficit on cancellation!	RECOMMENDED ARCHITECTURE

Banking Sweep Hours & Bank Holidays

Razorpay initiates nodal bank sweeps in automated batches (04:00 AM, 11:00 AM, 16:00 PM). If Friday is an RBI bank holiday, settlement is deferred to Monday, leaving the ₹1,335 inside the gateway buffer through Saturday.

Architectural Value of Escrow Hold

Holding vendor transfers until match kickoff creates a natural escrow buffer. The turf owner is assured payment will release automatically, while Turfify retains 100% liquidity to refund if weather or emergencies occur.

The Saturday Zero-Balance Problem: How Razorpay Handles It

When an authorized cancellation occurs and your Razorpay gateway balance is strictly **₹0.00** (because the payout already settled into your bank on Friday), here is the exact technical execution path.

RAIL A: AUTOMATIC OFFSET

Negative Merchant Balance

Razorpay permits the refund to proceed even with ₹0 funds by driving the account balance into the negative (e.g., **-₹1,500.00**).

Auto-Recovery: Upcoming bookings on Saturday afternoon automatically replenish this deficit. Zero penalties, no manual bank transfers needed.

RAIL B: QUEUED STATE

Queued / Pending Refund

If the merchant account has reached its negative balance tolerance limit, Razorpay places the refund in an active **"Queued"** queue.

Execution: The very instant the next customer pays for a turf slot, Razorpay intercepts the incoming cash flow and clears the queued refund.

RAIL C: MANUAL PRE-FUND

Instant Dashboard Top-Up

If there are no anticipated weekend transactions, the merchant can instantly top up their Razorpay balance directly via NetBanking, UPI, or NEFT.

Operation: Navigate to *Razorpay Dashboard → Balance → Add Funds*. Funds reflect instantly to satisfy pending refunds.

The Negative Balance Self-Healing Flowchart

STEP 1 (SATURDAY 10:00)
Refund Approved
Balance: **-₹1,500.00**

→

STEP 2 (SATURDAY 13:00)
New Booking (+₹1,500)
Incoming Player Checkout

→

STEP 3 (SATURDAY 13:01)
Auto-Reconciliation
Balance Restored to **₹0.00**

→

STEP 4 (SUNDAY 04:00 AM)
Net Payout Resumes
Fresh Profits Disbursed

Reversing the Turf Owner's Split via Razorpay Route

When an eligible refund occurs, the platform must never bear the turf owner's portion. Razorpay Route provides the `reverse_all: 1` API instruction to automatically claw back the transfer:

```
// Invoking an Idempotent Refund with Automated Route Transfer Reversal
const refund = await razorpay.payments.refund(paymentId, {
  amount: refundAmountInPaise,      // Full or partial refund amount
  reverse_all: 1,                   // Automatically reverses the turf owner's 90% transfer!
  notes: { booking_id: "TRF_BK_8921", reason: "Customer initiated cancellation" }
});
```

CRUCIAL BUSINESS REALITY

Should a Saturday Cancellation Even Get a 100% Refund?

NO! A customer who books on Wednesday for Saturday has blocked prime venue inventory for 3 days.

If they cancel on Saturday morning (< 6 hours before kickoff), standard industry practice and Turfify's Master Policy classify this as **Strictly Non-Refundable (0% Refund)**. The turf owner keeps their 85% payout because the slot cannot be re-booked on short notice. Therefore, the zero-balance refund issue is prevented by business logic!

Turfify Master Dynamic Cancellation Matrix

To balance customer fairness with venue financial security, Turfify implements a multi-tier cancellation protocol. Same-day cancellations protect the owner, while advance cancellations provide generous compensation.

TIMEFRAME	STATUS	PLAYER REFUND OUTCOME	TURF OWNER COMPENSATION
Tier 1: > 24 Hours Notice E.g., Thursday cancellation	GENEROUS	**90% Bank Refund** *OR 100% Turfify Wallet Credits* (10% retained for gateway fee)	**₹0 Payout** Slot is immediately unlocked and re-published on app for other players.
Tier 2: 6 to 24 Hours Notice E.g., Friday night cancellation	SHORT NOTICE	**50% Bank Refund** *OR 60% Turfify Wallet Credits* (50% forfeited penalty)	**40% Slot Compensation** Owner receives ₹600 as liquidated damages for late abandonment.
Tier 3: < 6 Hours / Same-Day E.g., Saturday for Saturday	STRICT NON-REF	**0% Refund (No Refund)** 100% booking value forfeited by user. Protects owner from empty turf!	**85% Payout to Owner** Owner retains ₹1,275. Turfify keeps 15% marketplace commission.
Tier 4: Ground Unplayable / Rain Force Majeure / Venue Fault	FORCE MAJEURE	**100% Immediate Full Refund** + ₹100 Bonus Wallet credit for inconvenience.	**₹0 Payout** Owner penalized on search ranking for venue unreadiness.

The Master Workaround: "Turfify Wallet Credits" (₹0 Cash Outflow)

The single most powerful tool for solving the zero-balance dilemma is offering **Turfify In-App Wallet Credits** instead of banking refunds:

1. Zero Cash Outflow
Zero money leaves your Razorpay account or your company bank account. Your cash balance stays 100% untouched.

2. Zero Gateway Fees
Bank refunds forfeit the initial 2% gateway processing fee. Wallet credits carry **₹0 in gateway charges**.

3. 100% Capital Retention
Player receives instant credit and immediately re-books another date, locking customer loyalty inside Turfify.

Comparative Analysis of Refund Rails

REFUND RAIL	SPEED TO CUSTOMER	TRANSACTION COST	IMPACT ON MERCHANT CASH
Original Source (Bank/UPI)	5 to 7 Business Days	Gateway retains ~2% fee	Cash exits gateway; requires balance or offset
Instant UPI (RazorpayX)	Under 60 Seconds	Additional ₹4–₹6 API payout fee	Cash exits bank instantly via IMPS
Turfify Wallet Credits (Recommended)	**Instant (0 Seconds)**	**₹0.00 (Zero Fee)**	**ZERO Outflow! Capital remains in your bank**

⭐ **The Win-Win Incentive: Why 82% of Players Choose Wallet Credits**
When presented with *"Receive 90% via Bank in 5–7 days OR Receive 100% Instantly in Turfify Wallet Credits"*, players overwhelmingly choose Wallet Credits. You preserve operating liquidity while providing an instantaneous, frictionless user experience.

Production Safeguards: Escrow Hold Protocol (`on_hold: 1`)

To permanently eliminate the zero-balance paradox at an architectural level, configure Razorpay Route to hold vendor transfers in escrow until the match is physically played:

```
// 1. Wednesday: Create Route Order with Settlement Held Until Saturday Match
const order = await razorpay.orders.create({
  amount: 150000, currency: "INR",
  transfers: [{
    account: turfOwnerAccountId,
    amount: 133500, currency: "INR",
    on_hold: 1,                          // FREEZES payout in escrow until released!
    on_hold_until: matchKickoffTimestamp  // Auto-releases after game ends
  }]
});

// 2. If Cancelled: Reverse Transfer Directly from Held Escrow (Zero Bank Clawback!)
await razorpay.transfers.reverse(transferId, { amount: ownerSharePaise });
```

Platform State Transition Matrix

STAGE	PAYMENT STATE	TURF OWNER SPLIT	ACTION ON CANCELLATION
Checkout Initiated	ORDER.CREATED	Unallocated	Order expires; zero deduction
Payment Captured	CAPTURED	Held in Route Escrow (`on_hold: 1`)	Transfer reversed in escrow; refund issued effortlessly
Match Finished	COMPLETED	Released (`on_hold: 0`)	Non-refundable (Service fulfilled)

Executive Client Cheat-Sheet: 5 Points to Tell Your Client

1. Razorpay Protects You with Auto-Offset
If your balance is ₹0, Razorpay executes the refund via a negative balance (-₹1,500) and recovers it from your next bookings without penalties or manual bank transfers.

2. Saturday Cancellations Are Non-Refundable
Clients who cancel on the day of the match (< 6 hours) are non-refundable under platform policy. You keep your 85% payout because the slot cannot be re-booked.

3. Route Automatically Pulls Back Splits
If an advance refund is issued, Razorpay Route reverses the turf owner's linked share automatically. The platform never pays refunds out-of-pocket.

4. Wallet Credits Keep 100% of the Cash
Offering 100% refunds as Turfify Wallet credits costs ₹0 in gateway fees and zero cash leaves your bank. The player simply reschedules their game.

Exact Message Script to Forward to Your Client:

"If a customer cancels on Saturday when your Razorpay balance is ₹0 because funds already settled to your bank on Friday, you are completely protected on three fronts: First, under venue policy, same-day cancellations are strictly non-refundable or offered as instant Turfify Wallet Credits (costing ₹0 out of pocket). Second, if an eligible refund is authorized, Razorpay natively covers it using a negative balance and automatically deducts it from upcoming weekend bookings. Third, using our Escrow Hold setting, payouts are held until match kickoff, guaranteeing zero cash deficits ever occur."